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PEKKA SIIRTOLA, University of Oulu, Oulu, North Ostrobothnia, Finland

HELI KOSKIMÄKI, University of Oulu, Oulu, North Ostrobothnia, Finland

JUHA RÖNING, University of Oulu, Oulu, North Ostrobothnia, Finland

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OpenHAR: A Matlab Toolbox for Easy Access to Publicly Open Human Activity Data Sets

Pekka Siirtola
Heli Koskimäki
Juha Röning

Biomimetics and Intelligent
Systems Group
University of Oulu
PO Box 4500
90014 University of Oulu
Finland

pekka.siirtola@oulu.fi
heli.koskimaki@oulu.fi
juha.roning@oulu.fi

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Abstract
This study introduces OpenHAR, a free Matlab toolbox to combine and unify publicly open data sets. It provides an easy access to accelerometer signals of ten publicly open human activity data sets. Data sets are easy to access as OpenHAR provides all the data sets in the same format. In addition, units, measurement range and labels are unified, as well as, body position IDs. Moreover, data sets with different sampling rates are unified using downsampling. What is more, data sets have been visually inspected to find visible errors, such as sensor in wrong orientation. OpenHAR improves re-usability of data sets by fixing these errors. Altogether OpenHAR contains over 65 million labeled data samples. This is equivalent to over 280 hours of data from 3D accelerometers. This includes data from 211 study subjects performing 17 daily human activities and wearing sensors in 14 different body positions.

Author Keywords
Human Activity Recognition; Accelerometer; Open Data

ACM Classification Keywords
I.5.4 [Pattern recognition]: Applications

Introduction
Inertial sensor based human activity recognition using wearable sensors and smartphone sensors has become

one of the most studied area of pattern recognition. One reason for this is that the results of activity recognition can be applied to many different types of applications. The most obvious one's include health and fitness monitoring. However, activity recognition can also be used for personalized advertising; smarthomes that anticipates the user's needs; and self-managing system that adapts to user's activities [7].

The first step to build a reliable human activity recognition model is to collect an extensive data set from the studied problem. Unfortunately, this can be very time consuming as this data not just needs to be collected but labeled as well. Luckily, it is not always necessary to collect a new data set as it is more and more common that data sets used in the previous activity recognition studies are made publicly available. Moreover, by combining multiple publicly open data sets, bigger data set can be build and bigger data set normally means more general and accurate recognition model. The problem is that it is not always that easy to combine data sets.

In our previous article [12], different publicly open human activity data sets were cross-validated. The recognition models were trained using one data set and tested using another to see how well models work when data for training and testing are collected in different environments and using different sensors. It was noted in the article that publicly open data sets are not always that easy to use and combine. For instance, it was noted that data sets are often stored in different formats, sensor orientation varies, units are not always the same, etc. Moreover, in [11] personalized human activity recognition models were studied and the experiments were based on publicly open data set containing data from ten study subjects. However, in the study, it was decided that data from one study subject was not used as apparently one subject had worn sensor in different orientation than others making this data non-uniform with other subjects data.

This study introduces OpenHAR, which is a free toolbox combining publicly open data sets. It provides an easy access to accelerometer signals of ten publicly open human activity data sets. Data sets are easy to access as OpenHAR provides all the data sets in the same format, units, measurement range and labels are unified, as well as, body position IDs. Moreover, data sets with different sampling rates are unified using downsampling. What is more, data sets have been visually inspected to find visible errors, such as sensors in wrong orientation. OpenHAR improves re-usability of data sets by fixing these errors. The study most similar to this is presented in [3], where a data set called AcctionNet collating six publicly open data sets was introduced. This data contains over 10 million labeled accelerometer samples samples from 13 activities. The data sets used in AcctionNet are partly the same as the ones used in OpenHAR. The main difference between AcctionNet and OpenHAR is that OpenHAR is not just a data set, it also provides tools to select only that part of data that is important to certain application.

OpenHAR

Collated data sets

OpenHAR combines ten publicly open data sets, which are listed in Table 1. Common with these data sets is that they all contain raw accelerometer data collected with reasonable sampling rate from activities of daily living. There is also other publicly open data sets available, but they were not included to this study as they do not fulfill our requirements: for instance, Reiss et al. [9] was not included to OpenHAR as data of it filtered and not raw. Moreover, SHL data set which is an excellent and extensive activity

data set by Gjoreski *et. al.* [5] was not included to OpenHAR as it is so huge compared to selected ten data sets that it would have a too dominant role in the combined data set.

According to Table 1, data sets of OpenHAR are not consistent which means that combining of the data sets is not as straightforward as it could be. Data sets are in multiple format and also data files are grouped in several different ways. In some cases, the whole data set is stored in one single file but often data are divided into multiple folders and files. In addition, when it comes to activity labels, both integers and strings were used as labels in the original data sets. Moreover, numerical labels did not have the same response in different data sets. In some cases, labels also had different meaning, for instance depending on the studied data set, *walking*, *walking upstairs* and *walking downstairs* had own labels but it was also possible that label *walking* included walking at flat level and walking at stairs. Another difference in the data sets is the used sampling frequency of acceleration data which varied from 40Hz to 200Hz. Moreover, accelerometer values differed in the provided value range and units. Visual mining of the data sets also showed some errors and non-uniformities from data sets. For instance, there are cases where sensor orientation of one study subject is not the same than for others.

Unifying data sets

The aim of the study was to unify the selected data sets, and therefore, provide an easy access to these data sets and combine them to get access to a bigger data set. The data sets presented in Table 1 comes in multiple file types and formats. In some cases, the whole data set is in one single file but often data are divided into multiple folders and files based on study subject ID, body position ID or activity labels. In fact, one of the main benefits of OpenHAR is that it provides code to load these without taking care of file formats, and the resulting data set has only one format.

Another problem in combining open data sets is that currently labels can be numeral or strings, and a number or string can have different meaning in different data sets. OpenHAR unifies these activity labels. It provides all the labels in numerical format and these labels have only one meaning. Activity labels used in OpenHAR are presented in Table 2. However, some of these activities are overlapping, which needs to be noted when OpenHAR data are used. For instance, [14] contained activity *idling*, which is a combination of sitting and standing, while in other data sets *sitting* and *standing* were considered as two separate activities. Similarly some data sets consider **walking**, *walking upstairs* and *walking downstairs* as separate activities and in some data sets these all three are considered as one activity called *walking*. The same goes to *elevator up* and *elevator down* activities, in some cases they are combined as *elevator*-activity (direction not defined). These subjects performed altogether 17 daily human activities, Table 2 although most data is from walking (19.9%), standing (15.6%) and sitting (13.1%) activities.

Depending on the purpose of the original article, sensor position differs between data sets. As the measured sensor values are greatly dependent on the body position of the sensor, each observation of OpenHAR has a cell defining from which body position the value has been measured. These positions and the position ID's are listed in Table 3. It is worth noting that some of these are overlapping. For instance, in some studies sensor position was defined as trouser's pocket, meaning that it can either left or right, while in some cases position was explicitly defined as left of right pocket. In addition, in some cases study subjects were allowed to decide the orientation of the sensor while in

Table 1: OpenHAR contains ten publicly open human activity data sets.

Data set ID	Author	File format	Frequency	Labels	Range and unit	Fixes
1	Banos *et. al.*[2]	.log	50Hz	numeral	$\pm 24m/s^2$	timestamp added
2	Ortiz *et. al.* [1]	.txt	50Hz	numeral	$\pm 2g$	timestamp added
3	Shoaib *et. al.* [10]	.csv	50Hz	strings	$\pm 20m/s^2$	subj. 8, belt: orientation fixed
4	Siirtola & Röning [14]	.txt	40Hz	numeral	$\pm 20m/s^2$	timestamp added
5	Stisen *et. al.* [15]	.csv	50 - 200Hz	strings	$\pm 40m/s^2$	sampling rates unified
6	USC-HAD [19]	.mat	100Hz	numeral	$\pm 6g$	timestamp added
7	UniMib-SHAR [8]	.mat	50Hz	numeral	$\pm 20m/s^2$	timestamp added
8	HuGaDB [4]	.txt	60Hz	numeral	± 32767	timestamp added
9	RealworldHAR [17]	.csv	50Hz	strings	$\pm 20m/s^2$	subj.8, chest: orientation fixed, subj.15, thigh: orientation fixed, subj.3, upperarm: orientation fixed, subj.3, waist: orientation fixed
10	MobiAct [18]	.csv	200Hz	strings	$\pm 20m/s^2$	-

some studies orientation was fixed. Moreover, some of the body positions were combined as they are so similar, for instance hip, waist and belt positions were combined as one. Most of the data is from hip (22.5%) and trouser's pocket (22.2%, including also data from thigh).

Another difference in the data sets is the used sampling frequency of acceleration data which varies from 40Hz to 200Hz. One feature of OpenHAR is that it unifies sampling rates using depending on which data sets are combined. Unifying is based on down-sampling by finding the greatest common divider of the sampling rates of the selected data sets. If all ten data sets are combined, the sampling rate of the resulting data set is 10Hz, which has been shown to be enough to reliably recognize activities [13, 6, 16].

Table 1 shows that unit of the measurement is either g (gravity) or m/s^2. Moreover, accelerometer differed in provided value range and units. OpenHAR converts all the units as m/s^2 to enable the joint usage of the data sets. In addition, the range of measurements is unified.

Visual mining of the data sets also showed some errors and non-uniformities from data sets. For instance, there are cases where sensor orientation of one study subject is not the same than for others. These were corrected by changing the common coordinate system to all data files within a data set. However, it should be noted that the orientation of a sensor is the same only within original data sets but when two original data sets are compared, the orientation of a sensor can differ. Nevertheless, this orientation issue can be solved for example using features extracted from magnitude signal, which is automatically calculated by OpenHAR using formula $\sqrt{(}\mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2)$, where **x**, **y** and **z** are the raw acceleration measurements from 3D accelerometer. Moreover, only a few original data set has timestamps, these were added to all data sets based on the sampling rate and by considering that it remains constant.

Altogether, OpenHAR contains over 65 million labeled data samples. This is equivalent to over 280 hours of data from 3D accelerometers. This includes data from 211 study subjects, see Figure 1. While the amount of data from each

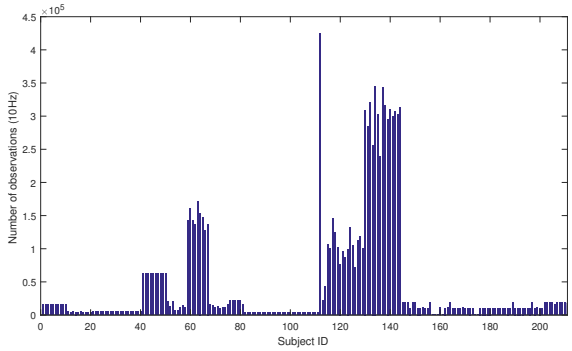

Figure 1: OpenHAR contains data from 211 study subjects.

study subject varies a lot between data set (minimum 2 minutes, maximum 710 minutes), on average there is 80 minutes of data from each subject.

Using OpenHAR

To use OpenHAR, the first step is to download the toolbox[1]. Next step is to download and unpack all ten original data sets (see ReadMe.txt -file to find from where to download them and where to unpack them). After this, everything is set. Note that OpenHAR requires Matlab.

Using OpenHAR is easy, only one command is needed to download all the data to *data*-file. (`[data, sampling_rate] = getOpenHAR()`). In addition, this command returns the sampling rate of data. In this case, all the data are downloaded and so the sampling rate would be 10Hz.

User can also load only wanted parts of the whole data set by three name-value -pairs arguments (`'datasets'`,

[1]OpenHAR is available at: http://www.oulu.fi/bisg/node/40364

Table 2: Data set includes accelerometer data from 17 activities. However, some of these are overlapping.

Activity ID	Activity	Amount of data
1	Standing	15.6%
2	Sitting	13.1%
3	Lying	8.0%
4	Idling (= sitting + standing)	0.4%
5	Walking	19.9%
6	Walking (inc. walking at stairs)	0.2%
7	Walking stairs up	10.3%
8	Walking stairs down	8.9%
9	Walking at stairs (inc. up and down)	0.2%
10	Running (inc. jogging)	10.4%
11	Biking	4.8%
12	Jumping	1.8%
13	Sitting in car	1.9%
14	Elevator up	1.1%
15	Elevator down	0.9%
16	Falling	0.7%
99	Null	1.9%

Table 3: OpenHAR includes data from 14 body positions. However, some of these are overlapping.

Position ID	Position	Amount of data
1	Hip (inc. belt and waist)	22.5%
2	Trouser's pocket, left (fixed orientation)	1.3%
3	Trouser's pocket, right (fixed orientation)	1.3%
4	Trouser's pocket, any (inc. thigh)	22.2%
5	Chest	7.0%
6	Wrist, any (inc. forearm)	10.1%
7	Upper arm	7.7%
8	Head	6.4%
9	Shin (inc. leg)	13.3%
10	Ankle	0.5%
11	Trouser's pocket, left (free orientation)	0.5%
12	Trouser's pocket, right (free orientation)	0.5%
13	Foot, left	3.4%
14	Foot, right	3.4%

`'activities'`, and `'positions'`). User can specify several or only one name and value pair argument in any order.

If the purpose is not to use data from all ten original data sets, wanted data sets can be specified as the comma-separated pair consisting of `'datasets'` and a vector containing th IDs of the wanted data sets, see Table 1 for dataset IDs. For example, command `[data, sampling_rate] = getOpenHAR('datasets', [1 3 5])` returns only measurements from original data sets corresponding to data set IDs 1 ([2]), 3 ([10]), and 5 ([15]). If all the selected data sets do not have the same sampling frequency, OpenHAR unifies them and return the sampling rate of the combined data set.

Similarly, if only some activities or body positions are of interest, wanted activities and body positions can be specified as the comma-separated pair consisting of `'activities'` and `'positions'`, and a vector containing the IDs of the wanted activities or body positions. IDs for these are listed in Tables 2 and 3. This means that user can select only data from some activities or body positions, or only data from some activities from selected body positions.

Each case, the code returns *data*-file. This file has nine columns of data in the following order: *data set ID, position ID, user ID, activity ID, timestamp, x-axis acceleration, y-axis acceleration, z-axis acceleration* and *magnitude acceleration*. File does not have a header.

Discussion and conclusion

This article presents OpenHAR, a Matlab toolbox combining ten publicly available human activity data set. The extra value provided by OpenHAR is that it provides easy access to these ten data sets. In fact, OpenHAR provides all the data sets in the same format. In addition, units, measurement range and labels are unified, as well as, body position IDs. Moreover, data sets with different sampling rates are unified using downsampling. What is more, data sets have been visually inspected to find visible errors, such as sensors in wrong orientation. OpenHAR improves re-usability of data sets by fixing these errors. With over 65 million labeled observation, 211 study subjects, 17 activities and 14 body position, OpenHAR is the most comprehensive accelerometer based human activity data set to date.

OpenHAR opens a lot of new possibilities to researchers and application developers. For instance, OpenHAR provides a great testbed to study deep learning methods and other data hungry classifiers. In addition, OpenHAR contains data from ten different data gathering protocols, which means that using OpenHAR is it possible to experiment how a model that is trained using data from one location and environment works when it is tested in other location. Thus, it can be used to test methods of transfer learning. In addition, OpenHAR contains data from 211 study subjects, which is more than any other data sets. The high number of study subjects is especially important when user-independent models are trained. Moreover, when used with traditional classifiers, more data usually means better and more accurate models. Most importantly, OpenHAR is publicly open, and therefore, studies based on it are replicable.

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